

Web Server Setup for WordPress on Raspberry Pi

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WordPress is a PHP and MySQL based open source blogging platform and content management system. It is a popular blogging system and includes features like threaded comments, galleries, revision histories, trash, custom post types, thousands of themes, widgets and plug-ins. This article describes the setting up of a web server for WordPress on Raspberry Pi.

Apache web server installation

Apache, a web server application notable for playing a key role in the initial growth of the World Wide Web, is used to serve web pages. Follow the procedure given below to install it on Raspberry Pi.

1. Install apache2 package by executing the following command (also

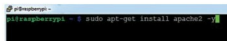


Fig. 1: Installing apache2 package



Fig. 2: Starting Apache server



Fig. 3: Stopping Apache server



Fig. 4: Browser default HTML page

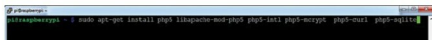


Fig. 5: Installation of all PHP packages

see Fig. 1):

```
$ sudo apt-get install apache2 -y
```

2. After this installation, Apache server will be running. Thereafter you can start or stop it by running the following commands (also see Figs 2 and 3):

```
$ sudo service apache2 start
```

```
$ sudo service apache2 stop
```

3. Apache server by default creates a test HTML file in web folder. This can be served when `http://localhost/` is browsed on the Raspberry Pi or `http://192.168.2.104` (IP address of the Raspberry Pi in our case, but can be different in your case) from any other computer on the network.

Browse the default web page and you will see the message shown in Fig. 4 appear on the screen.

4. Change the default web page as per your requirement. The default web page is an HTML file on the file system and is located at `/var/www/index.html`. Execute the following command on terminal to navigate and have a look on the file:

```
$ cd /var/www
$ ls -al
```

You will see the following:

```
total 12
drwxr-xr-x 2 root root 4096 Mar 2
03:15 .
drwxr-xr-x 12 root root 4096 Mar 2
03:15 ..
-rw-r--r-- 1 root root 177 Mar 2
03:16 index.html
```

This shows that there is one file called `index.html` in `/var/www/`. The `.` (dot) at the end of line 2 refers to the directory itself `/var/www/` and the `..` at the end of line 3 refers to the parent directory `/var/`.



Meaning of each column in the previous file is:

1. File or directory permissions
2. Number of files in the directory (1 in case of a file)
3. User who owns directory or file
4. Group which owns the directory or file
5. File size
6. Date and time of last modification

In order to edit the directory and file, you should log in as root user. Edit the `index.html` file using either of the following commands.

```
$ sudo chown pi: index.html
```

```
$ sudo nano index.html
```

Try editing/changing this file and refreshing the browser to see the web page changes.

Next step is to install PHP and MySQL, which are used to create your own site and publish your own content dynamically, without knowing how to program those pages.

PHP installation

Install PHP by executing the following command (also see Fig. 5):

```
$ sudo apt-get install php5 libap
ache-mod-php5 php5-intl php5-mcrypt
php5-curl php5-sqlite
```

To check whether PHP has been installed properly, move `index.html` file to `index.php` using following command (also see Fig. 6):

```
$ sudo mv index.html index.php
```

Now edit the file:

```
$ sudo nano index.php
```

Put following text in `index.php` after opening the nano editor:

```
<?php echo "hello world";
```

Save and then refresh the browser. You should see 'hello world' on your